



Team ID: 21
Team Name: RAG Busters
EDU-RAG: Discourse-Aware RAG for FEVER



Abstract

We investigate whether *discourse-aware* segmentation into Elementary Discourse Units (EDUs) improves Retrieval-Augmented Generation across two benchmarks: **FEVER** (fact verification) and **SQuAD** (open-domain QA). In **Phase 1**, we train EDU boundary detectors on DISRPT-2025 treebanks (BERT + MLP head, **0.9070 F1**). In **Phase 2**, we benchmark four chunking strategies using all-MiniLM-L6-v2 + Qdrant across both datasets. On SQuAD, EDU chunking yields higher **Accuracy@5 (0.892 vs. 0.853)** and **Recall@5 (0.887 vs. 0.929)**, but lower **Precision@1 (0.669 vs. 0.745)** and **MRR (0.767 vs. 0.826)** compared to sentence chunking. In **Phase 3**, an end-to-end RAG pipeline with Gemma-2B reveals that chunking strategy must be matched to corpus structure and evaluation granularity — discourse-aware chunking excels for broad recall, while sentence chunking dominates top-1 precision tasks.

Goal

To evaluate whether discourse-aware EDU-based chunking provides superior retrieval performance compared to sentence-level chunking across two corpora of different annotation granularity: **FEVER** (sentence-level ground truth) and **SQuAD** (passage-level QA). We assess both retrieval effectiveness (Recall@k, Accuracy@k, MRR, Precision@1) and downstream generation quality, testing the hypothesis that *chunking strategy must match corpus structure* rather than being treated as a universal design choice.

Dataset

DISRPT-2025 EDUs

- 7 English discourse corpora (GUM, SciDTB, STS, etc.), **1,681** docs total (**1,247** train / **189** dev / **245** test) used to train our EDU boundary detector.
- Strong class imbalance ($\approx$ 1:8 EDU starts vs. continues); EDU-start tokens are upweighted $\times 4$ in the loss.

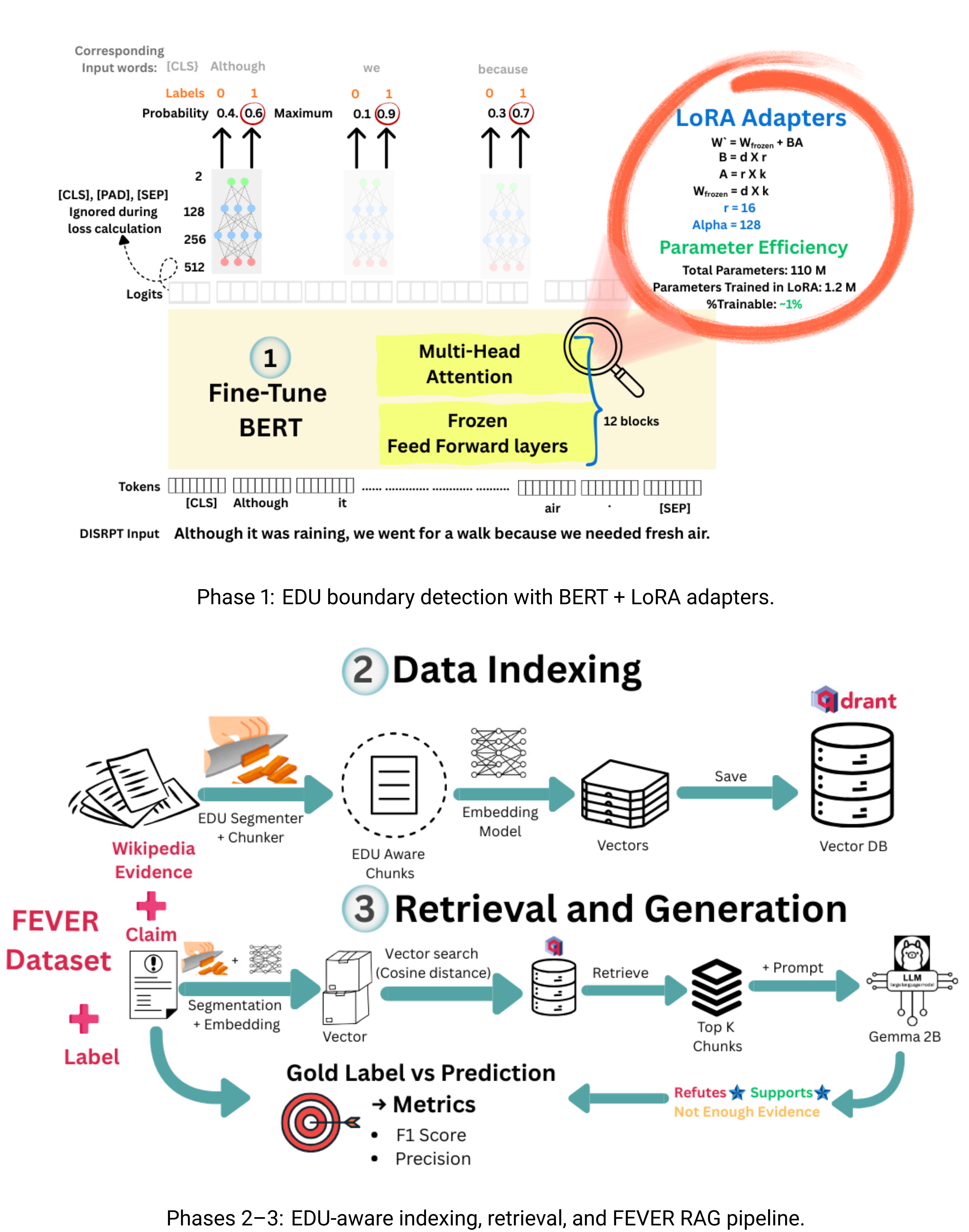
FEVER Dataset (reduced)

- Wikipedia fact-checking dataset of claims with supporting/refuting evidence.
- We index **55,407** evidence sentences and evaluate on **2,000** dev + **2,000** test claims from a reduced split with $\sim$ 5,000 relevant evidences.
- Labels are nearly balanced: $\sim$ **33%** SUPPORTS, **33%** REFUTES, **33%** NOT ENOUGH INFO.
- Ground truth is **sentence-level**, naturally favouring sentence chunking.

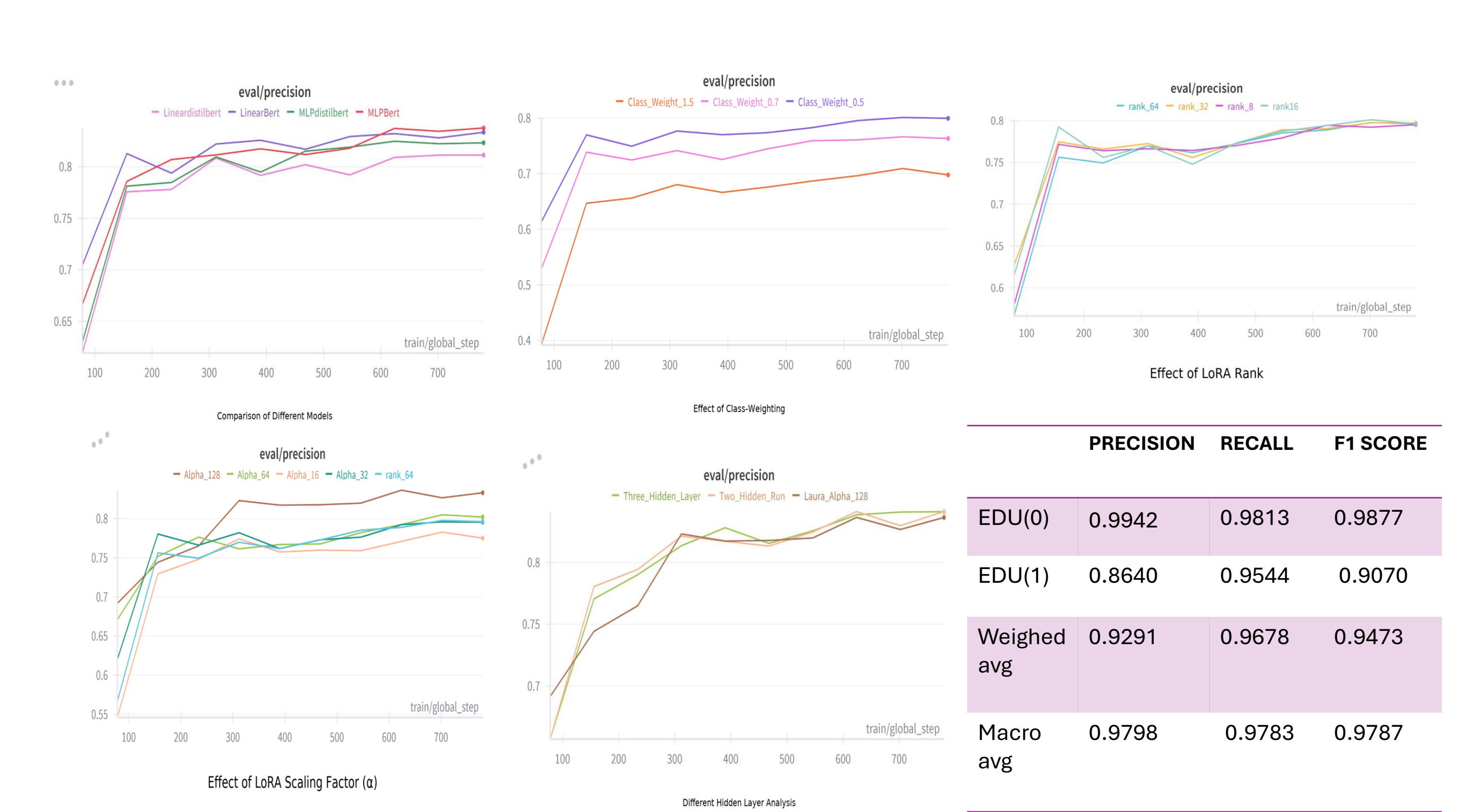
SQuAD Dataset

- Stanford Question Answering Dataset: **5,928** questions over Wikipedia passages.
- Answers span longer, multi-sentence contexts — ground truth is **passage-level**, making this a better test bed for discourse-aware chunking.
- Each question has exactly one relevant document; evaluation uses Precision@k, Recall@k, Accuracy@k, and MRR across $k \in \{1, 3, 5, 10, 20\}$.

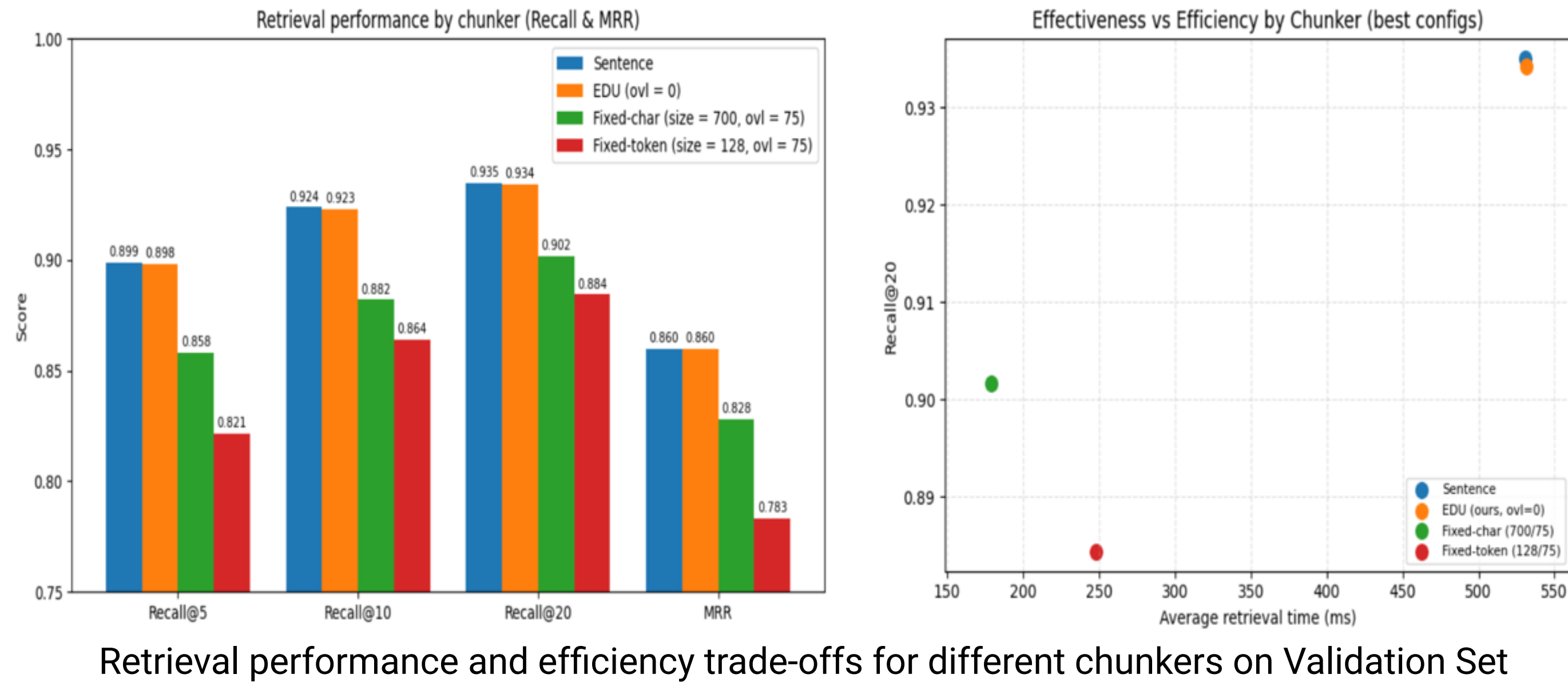
Methodology



Results



Precision curves for EDU boundary models under different architectures, class weights, LoRA ranks, and scaling factors.



Retrieval performance and efficiency trade-offs for different chunkers on Validation Set

Chunker	MRR	Recall@5	Precision@5	Accuracy	F1 Score	Precision	Recall
Custom-EDU	0.8569	0.8878	0.1986	0.4030	0.3513	0.4950	0.4030
Sentence	0.8570	0.8971	0.2006	0.4170	0.3653	0.5163	0.4170
Fixed-Char	0.8371	0.8747	0.1961	0.4020	0.3510	0.4738	0.4020

Downstream FEVER classification performance for each chunker						
Chunker	MRR	P@1	Acc@5	Rec@5	Acc@20	Rec@20
Sentence	0.826	0.745	0.853	0.929	0.933	0.974
EDU based	0.767	0.669	0.892	0.887	0.969	0.966

SQuAD Retrieval Results: EDU vs. Sentence Chunking

Bold = best per column. EDU chunking improves Accuracy@5 (+3.9%) and Accuracy@20 (+3.6%) while sentence chunking leads on top-1 precision and MRR.

Analysis

Phase 1 – EDU Segmentation.

False-positive boundaries fragment discourse units, degrading chunk coherence and downstream retrieval. Missing boundaries are less harmful since chunks combine 3–5 EDUs and stay coherent despite occasional gaps. Class-weighted loss up-weights rare EDU-start labels, preventing a collapse to “no boundaries” while keeping a strong precision focus.

Phase 2 – Retrieval Strategy.

We prioritise **Recall@k** because missing evidence is fatal — the LLM cannot classify correctly without the right chunks, whereas extra noise is often recoverable. **MRR** captures how early the first relevant chunk appears; higher MRR means better evidence prioritisation. On FEVER, sentence and EDU chunkers reach comparable Recall@20 (0.935 vs. 0.926), but EDU yields lower MRR and Precision@1: sentence-level gold annotations cause sub-sentence EDU splits to straddle span boundaries. On SQuAD, whose passage-level answers better reward discourse-aware splits, EDU improves Accuracy@5 (+3.9%) and runs at comparable speed, yet sentence chunking still leads on MRR (0.826 vs. 0.767) and Precision@1 (0.745 vs. 0.669) due to segmentation noise at scale (2.33 EDUs/sentence; 679K boundaries).

Phase 3 – Pipeline & Conclusion.

Small retrieval gaps amplify sharply downstream. Small retrieval drops can amplify downstream, as observed in our FEVER setup a $\sim$ 0.8% recall drop translates to Accuracy $-$ 5.3% and F1 $-$ 5.8%: encyclopedic Wikipedia sentences are already coherent semantic units, so EDU splitting adds errors without benefit and inflates false NOT ENOUGH INFO predictions. On SQuAD, EDU’s broader recall pays off for multi-hop questions where answers span multiple discourse units. *Chunking strategy must match corpus annotation granularity — sentence boundaries win when gold spans are sentence-aligned; discourse-aware chunking unlocks gains where context is richer and annotation is coarser.*

Limitations and Future Work

Our results highlight that the effectiveness of discourse-aware chunking is highly dependent on dataset characteristics and annotation granularity. In particular, for datasets such as FEVER, where evidence is annotated at the sentence level, simple sentence-based chunking remains a strong baseline and often outperforms finer-grained EDU segmentation due to better alignment with ground-truth spans.

In contrast, discourse-aware chunking shows greater promise for complex, long-form reasoning tasks, where relevant information may span multiple clauses or discourse units and is not confined to sentence boundaries.

Future work should therefore evaluate EDU-based segmentation on corpora with longer and more syntactically complex sentences, where discourse structure is more critical. Additionally, expanding training to more diverse discourse datasets and exploring full-model fine-tuning (beyond a linear head) may improve segmentation quality. Further directions include tuning EDU window sizes (e.g., 3–7 EDUs) to better balance retrieval effectiveness and storage costs, and exploring techniques like few-shot prompting, chain-of-thought reasoning, adaptive retrieval (ARR), and larger models.